Why an alphabet rather than words? An iterated-learning model of combinatorial structure in sperm whale codas

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github.com/loai-elhaj/whale-alphabet-emergence

Status. Undergraduate project, not peer-reviewed. The model is symbolic (no audio) and tests the *plausibility of mechanisms*, not what happened in real sperm whales. Every number below is produced by the code in the repository (seeded and deterministic).

1. Summary

Sharma et al. (2024) showed that sperm whale codas combine rhythm, tempo, rubato and ornamentation, which yields a much larger signal inventory than the 21 coda types described previously. We ask which cultural-transmission pressures are *sufficient* for such combinatorial ("alphabet-like") structure to emerge in a minimal iterated-learning model, compared with a repertoire of unrelated holistic signals ("words"). Three mechanisms were tested:

	Mechanism	Outcome in the model
H1	Transmission bottleneck	**Supported.** Compositionality rises from 0.00 (no bottleneck) to 0.64 (bottleneck 0.5) and plateaus near 0.70–0.72.
H2	Selection for communication precision	**Weak / inconclusive.** Precision rises (0.88 to 0.97) but at 40 generations compositionality does not change detectably; an exploratory 150-generation run shows a small effect (+0.026, p = 0.022, uncorrected).
H3	Homophily (learning mostly from one's own clan)	**Supported within the model.** Weaker homophily makes clan-specific structure rarer and compositionality lower; the transition lies between homophily 0.8 and 0.5.

H1 reproduces a well-known result of the iterated-learning literature (Kirby et al., 2008); the

contribution here is applying it to a clan/unit/individual meaning space inspired by sperm whale social structure, and exploring H2 and H3 in that setting.

2. Biological background and what the model borrows from it

Codas. Codas are stereotyped sequences of clicks. Sharma et al. (2024) analysed 8,719 codas from Eastern Caribbean whales (Dominica Sperm Whale Project): codas cluster into 18 rhythm types and 5 tempo types, with rubato (gradual duration variation across successive codas) and ornamentation (an extra final click); at least 143 rhythm/tempo combinations are frequently realised, versus 21 coda types described before. The paper shows structure, not meaning, and includes no playback experiments.

Identity cues. Gero, Whitehead and Rendell (2016) studied 4,116 codas from nine Caribbean social units and found identity cues at the level of the individual, the unit and the vocal clan. Units numbered 4–12 individuals (mean 8) and all belonged to one vocal clan; two coda types ("1+1+3" and "5R1") made up 65 % of codas; four units each used a distinctive 4-click type. Individual identification from "5R1" worked within units (73–83 % in the cases tested) but poorly when units were pooled (46 %). In the Pacific, Rendell and Whitehead (2003) described over 70 coda types but only five vocal clans (as cited by Gero et al., 2016).

What the model borrows. The hierarchy *clan* > *unit* > *individual* gives the structure of the meaning space. This is a **modelling assumption**: it treats identity as what signals encode, whereas Sharma et al. do not establish what codas mean. The sizes used (6 clans × 5 units × 11 individuals = 330 meanings) are illustrative and **not** measured values: the Pacific literature reports five clans, Caribbean units have 4–12 members (mean 8), and the number of units per clan is arbitrary.

3. Model

Meanings and signals. A meaning is a triple (clan, unit, individual) with (6, 5, 11) values. A signal is a 6-letter string over a 6-letter alphabet, made of three 2-letter slots, one per dimension.

Agents. Each agent has (i) a *lexicon*, the exact (meaning → signal) pairs it observed, and (ii) a table of *chunks* mapping a (dimension, value) to a 2-letter slot. After observing examples, a chunk is induced for a (dimension, value) when more than 50 % of the observed examples sharing that value carry the same slot (majority rule; at least 2 examples).

Production. For a known meaning the agent recalls the lexicon entry, copied with 5 % per-letter noise. For an unknown meaning it concatenates the chunks of the meaning's three values; a missing chunk is invented once and kept thereafter.

Generations. Generation 0 has a random lexicon. Each subsequent agent observes a random fraction (the *bottleneck fraction*) of the previous agent's 330 signals, learns, then produces the full language. We simulate 40 generations; results are taken at the last generation.

Compositionality is measured by *topographic similarity*: the Spearman rank correlation, over all pairs of meanings, between the Hamming distance of the meanings and the Levenshtein distance of their signals (average ranks for ties). A random language gives about 0; a perfectly compositional one approaches 1. **Chunks learned** is counted right after learning, before production, so that chunks invented at production time do not inflate it (maximum 22 = 6 + 5 + 11).

Statistics. Final-generation scores are compared across independent seeds with one-sided Mann–Whitney U tests, with Bonferroni correction for multiple comparisons.

4. H1 — Transmission bottleneck

Prediction. A tighter bottleneck forces generalisation, so topographic similarity should increase as the bottleneck fraction decreases. **Design.** Bottleneck fractions 1.0, 0.7, 0.5, 0.3, 0.15; 20 seeds each.

Bottleneck	Topographic similarity (mean ± sd)	Chunks learned (of 22)
1.0 (none)	−0.001 ± 0.004	0.0
0.7	0.302 ± 0.065	10.6
0.5	0.636 ± 0.030	22.0
0.3	0.697 ± 0.035	21.9
0.15	0.717 ± 0.027	21.2

Consecutive comparisons (one-sided, Bonferroni $\alpha = 0.0125$): 0.7 > 1.0, $p = 3.4\times10^{-8}$; 0.5 > 0.7, $p = 3.4\times10^{-8}$; 0.3 > 0.5, $p = 3.8\times10^{-6}$; 0.15 > 0.3, $p = 0.034$ (not significant after correction).

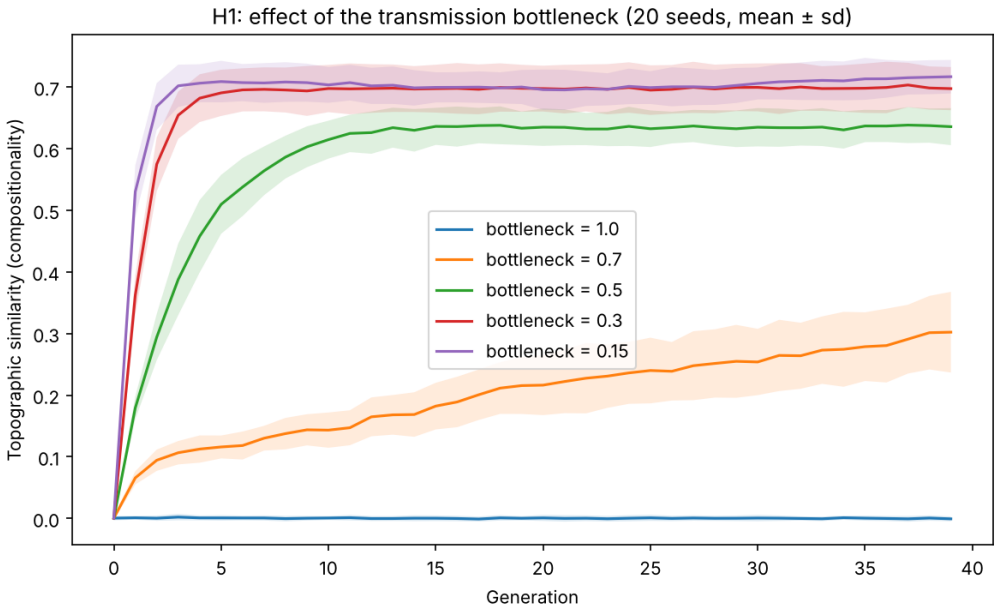


Figure 1: H1: topographic similarity across generations for each bottleneck (mean ± sd over 20 seeds).

Reading. Without a bottleneck, every signal is simply memorised and no structure appears. With a bottleneck, the learner must reconstruct unseen meanings from chunks, and only compositional signals survive transmission. The gain saturates once nearly all 22 chunks are learned. **H1 is supported**, with the caveat that this is the classic iterated-learning effect.

5. H2 — Selection for communication precision

Prediction. If agents whose signals are less ambiguous are favoured, compositionality should increase beyond the bottleneck effect alone. **Design.** At each generation, N independent candidate learners are produced (each from its own random sample of the bottleneck); only the one with the highest *communication precision* (fraction of the 330 meanings whose signal is shared

with no other meaning) is kept. N ∈ {1, 3, 6, 10}; N = 1 is the reference (identical to H1 at 0.7). Bottleneck 0.7 (the only tested value not yet at ceiling); 20 seeds.

Candidates	Topographic similarity	Chunks (of 22)	Precision	p vs. N = 1 (α = 0.0167)
1 (reference)	0.302 ± 0.065	10.6	0.879	—
3	0.321 ± 0.059	10.9	0.943	0.212
6	0.320 ± 0.062	11.3	0.970	0.182
10	0.317 ± 0.057	10.5	0.973	0.254

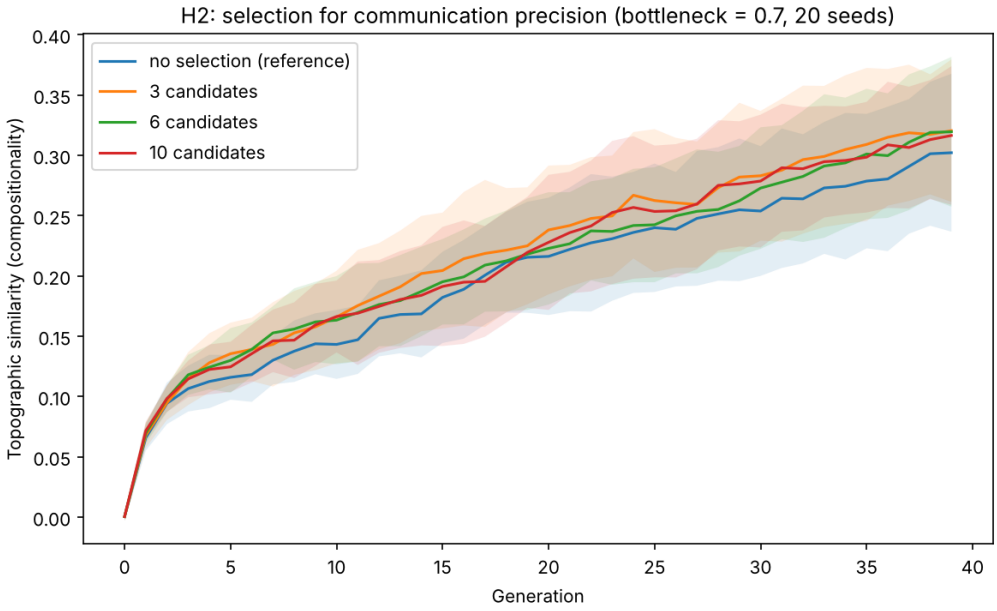


Figure 2: H2: topographic similarity across generations by selection strength (mean ± sd over 20 seeds).

Reading. The selection acts as intended (precision rises from 0.88 to 0.97) but at generation 40 compositionality does not detectably change. This is consistent with the idea that being unambiguous and being compositional are different properties: a language can be precise simply because its signals are arbitrary and different. However, Figure 2 shows that at bottleneck 0.7 compositionality is still rising at generation 40, so a selection effect could need longer to appear.

Supplementary long run (exploratory, run after seeing the result above). We compared 1 vs. 10 candidates over 150 generations (20 seeds each; `run_h2_long.py`): topographic similarity at the last generation was 0.495 ± 0.043 (no selection) vs. 0.521 ± 0.040 (10 candidates), one-sided Mann–Whitney p = 0.022. This is a small effect (about +0.03), obtained from a single follow-up comparison chosen after the main result, so it is not corrected for multiple testing and should be treated as suggestive. **Verdict for H2: no detectable effect at 40 generations; at most a small effect in longer runs.** The main takeaway is that selection for precision is far weaker than the bottleneck (H1) as a driver of compositionality in this model.

6. H3 — Homophily and clan differentiation

Prediction. If learners draw their examples mostly from their own clan, each clan should stabilise its own chunks (differentiation); as examples increasingly come from other clans, whose

chunks differ, structure should become harder to maintain. **Design.** Six clan lineages evolve in parallel, each over its own 55 meanings (5 units × 11 individuals). For every learning example, with probability *homophily* it comes from the learner's own clan; otherwise from a random other clan (same unit/individual pair, signal produced by that other clan). Homophily ∈ {1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.3, 0.2}; bottleneck 0.5; 30 seeds per condition.

Three measures: (1) mean topographic similarity across the six clans; (2) a *differentiation index*, defined only for (unit/individual) values for which at least 4 of 6 clans learned a chunk, as the fraction of distinct chunks among those clans (1 = every clan uses its own chunk); (3) the *probability that structure forms*, the fraction of seeds in which at least one value meets that criterion (defined for all seeds).

Homophily	Topographic similarity	Differentiation index	Seeds with index defined (of 30)	P(structure forms)
1.0	0.555	0.939	30	1.00
0.9	0.435	0.926	30	1.00
0.8	0.353	0.905	29	0.97
0.7	0.255	0.822	22	0.73
0.6	0.218	0.880	15	0.50
0.5	0.195	0.559	10	0.33
0.3	0.168	0.333	3	0.10
0.2	0.161	0.375	4	0.13

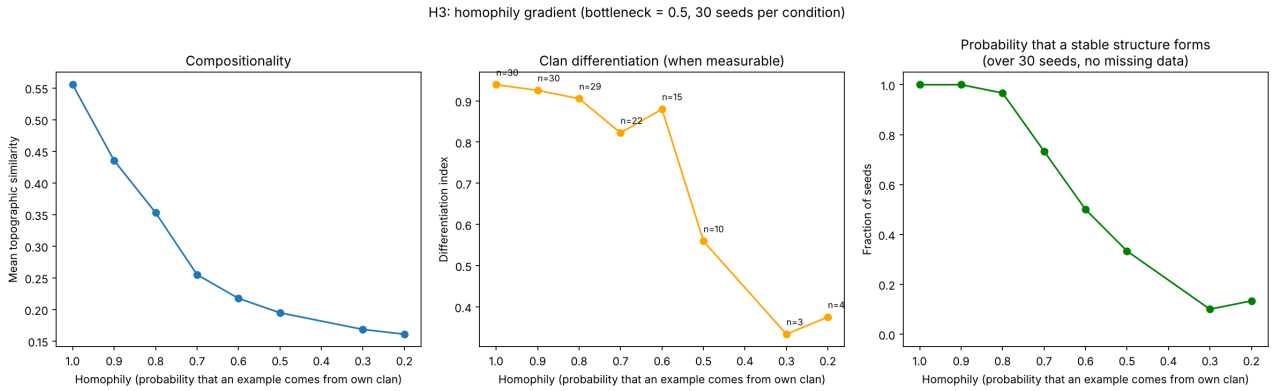


Figure 3: H3: three measures as a function of homophily (30 seeds per condition).

Reading. Compositionality falls monotonically as homophily weakens, and the probability that clan-specific structure forms crosses 50 % between homophily 0.7 and 0.6. The differentiation index is only trustworthy while enough seeds qualify (down to homophily 0.6); below that it rests on 3–10 seeds and is selected on the very condition (structure forming) that it is meant to measure, so it is indicative only. **H3 is supported within the model.** Mechanistically, clans invent different chunks, so mixing examples across clans yields contradictory evidence and the majority rule fails to consolidate chunks. This is a property of the model's learning rule and should not be read as direct evidence about real sperm whales. Note also that H3 scores are computed on 55 meanings per clan (one fixed dimension) and are **not directly comparable** to the 330-meaning scores of H1 and H2.

7. Validation and design pitfalls found along the way

Documenting these is part of the result, because each one changed numbers.

1. **Strict unanimity learned nothing.** The first chunk-induction rule (all examples must agree) produced zero chunks under 5 % copy noise; it was replaced by a majority rule.
2. **Chunk priority caused a regression.** An early version gave chunks priority over the lexicon during production, which made all 22 chunks "learned" regardless of condition. It was removed; a reference check (bottleneck 0.7, seed 0 → 10 chunks) guards against its return. An accidental reintroduction later invalidated one H2/H3 run, which was discarded after the check failed.
3. **Measuring chunks too late.** Chunks counted after production include invented ones. Both the chunk count (H1/H2) and the clan differentiation (H3) are now snapshotted right after learning.
4. **Total isolation was an uninformative H3 test.** With no information shared between clans, independent lineages diverge almost surely (two random 2-letter slots coincide with probability 1/36 for a 6-letter alphabet); replaced by the homophily gradient.
5. **Tied ranks in Spearman correlation.** The first implementation ranked with a bare `argsort`. Hamming distances take only 4 values and Levenshtein distances 7, so ties are massive, and arbitrary tie-breaking both biased scores upward (random-language baseline of 0.04 instead of 0) and made results depend on the machine (the same seed gave different scores on two computers). It was replaced by average ranks. After the fix all chunk counts, precisions, differentiation indices and formation probabilities were identical to those obtained before; only topographic similarities changed, and no conclusion changed.

8. Limitations

- **Symbolic signals.** No acoustics; the 3-slot segmentation is imposed rather than discovered. A model closer to the data would have to learn the segmentation (and the dimensions of rhythm, tempo, rubato, ornamentation) from audio.
- **Assumed meaning space.** Identity (clan/unit/individual) is assumed to be what codas encode; the actual information content is unknown. Sizes are illustrative.
- **Single conditions.** H2 was tested at bottleneck 0.7 and H3 at 0.5; no full factorial cross.
- **Simplified mechanisms.** Selection is on strict signal uniqueness only; homophily is a binary own-clan/other-clan choice with no graded social proximity; no genetic component (codas are culturally transmitted).
- **H2 is only weakly assessed.** No effect at 40 generations (n = 20); a small effect in a post hoc 150-generation run. Other forms of selection were not tested, and a longer, pre-planned design would be needed to settle it.
- **No quantitative match with real data.** We deliberately do not claim that model quantities (e.g. 330 meanings from 22 chunks) correspond to empirical ones (e.g. ≥143 combinations from 18 rhythms and 5 tempos).

9. Reproducibility

Python 3, `numpy`, `scipy`, `matplotlib`. Scripts `run_h1_bottleneck.py`, `run_h2_selection.py`, `run_h3_homophily.py` (and `run_h2_long.py`) regenerate every table and figure (about 7 minutes in total on a laptop; `run_h2_long.py` adds about 3 minutes); `make_summary_figure.py` rebuilds the summary figure. Seeds are fixed (0–19 for H1/H2, 0–29 for H3).

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